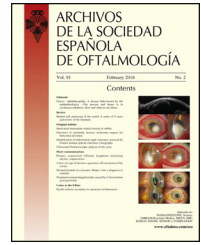




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Short communication

Bilateral parainfectious papillitis superimposed on papillary pseudoedema by optic nerve drusen



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ARTICLE INFO

Article history:

Received 20 January 2025

Accepted 21 April 2025

Available online 19 July 2025

Keywords:

Papillitis

Optic neuritis

Bilateral

Parainfectious

Sex transmitted disease

Chlamydia trachomatis

ABSTRACT

Papillary drusen are deposits of calcified hyaline material at the level of the optic nerve head. They are considered the most frequent cause of pseudopapilledema, so in clinical practice it is essential to identify and differentiate them from true optic disc oedema.

Parainfectious optic neuritis is an inflammatory optic neuropathy, usually bilateral and acute, which occurs between 1 and 3 weeks after an infectious disease. It usually occurs as a papillitis, since it is usually located in the head of the optic nerve causing optic disc oedema.

In this article we present an unusual case of bilateral papillitis superimposed on papillary drusen in a young male with *Chlamydia trachomatis* prostatitis. After an exhaustive etiological study, it was concluded that, in this patient, papillitis was associated with his sexually transmitted disease.

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Papilitis parainfecciosa bilateral superpuesta a pseudoedema de papila por drusas del nervio óptico

RESUMEN

Las drusas de papila son depósitos de material hialino calcificado a nivel de la cabeza del nervio óptico. Se las considera la causa más frecuente de pseudopapiledema, por lo que en la práctica clínica es fundamental su identificación y diferenciación de un verdadero edema de papila.

La neuritis óptica parainfecciosa es una neuropatía óptica inflamatoria, habitualmente bilateral y aguda, que se presenta entre 1 y 3 semanas después de una enfermedad infecciosa. Por lo general cursa como una papilitis, ya que suele localizarse en la cabeza del nervio óptico provocando un edema de papila.

Palabras clave:

Papilitis

Neuritis óptica

Bilateral

Parainfecciosa

Enfermedad de transmisión sexual

Chlamydia trachomatis

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<https://doi.org/10.1016/j.oftale.2025.07.004>

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En este artículo presentamos un inusual caso de papilitis bilateral superpuesta a drusas de papila en un varón joven con una prostatitis por *Chlamydia trachomatis*. Tras un exhaustivo estudio etiológico se concluye que, en este paciente, la papilitis estaba asociada a su infección de transmisión sexual.

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Introduction^{1–3}

Parainfectious optic neuritis is an inflammatory optic neuropathy that typically appears between 1 and 3 weeks after an infectious illness, usually viral and less commonly bacterial. Although it can occur at any age, it's more common in childhood. Clinically, it often presents as an acute bilateral optic neuropathy with optic disc edema (papillitis), sometimes accompanied by lipid exudates in the macular area (neuroretinitis). The impact on visual acuity (VA), dyschromatopsia, and visual field (VF) abnormalities varies widely, ranging from subtle changes to severe loss. It's a rare condition for which diagnosis requires ruling out other causes of optic disc edema. Generally, once the underlying infection resolves, the prognosis is usually excellent with or without anti-inflammatory treatment.

Optic disc drusen are deposits of calcified hyaline material at the optic nerve head (ONH). They are considered the most common cause of pseudopapilledema. Their prevalence varies between 0.4% and 2.4%, with a predominance in women and Caucasian individuals, and a strong familial tendency. Their etiopathogenesis seems to be related to an alteration in axonal metabolism and slowed axoplasmic flow, likely linked to an anomaly in optic disc development. Although considered benign processes, most patients develop VF (visual field) defects in adulthood.

In literature, we find cases of pseudopapilledema due to optic disc drusen that temporarily develop superimposed true optic disc edema, either due to ischemic optic neuropathy⁴ or papilledema in the context of intracranial hypertension (ICH).⁵ We present an unusual case of bilateral optic disc edema due to parainfectious papillitis superimposed on bilateral pseudopapilledema following optic disc drusen.

Case report

A 32-year-old man, with no relevant personal or family history, was on empiric treatment with cefixime due to a 10-day history of prostatitis. He presented to ophthalmology emergency with a 3-day history of subjective bilateral VA loss and an inferior VF defect in the right eye (OD), along with occipital headache, fever, and general malaise. Blood pressure, neurological examination, and blood tests all turned out normal, except for a C-reactive protein (CRP) of 104 mg/L attributable to his infectious condition.

VA with best correction was 0.7 in OD and 1 in the left eye (OS). Extrinsic and intrinsic ocular motility was normal,

and confrontation visual field testing was normal in OS but revealed an inferior altitudinal defect reaching the central VF in OD.

The anterior segment and vitreous chamber showed no abnormalities, and intraocular pressure was 15 mmHg in both eyes (BE). Funduscopic evaluation ruled out retinal abnormalities but showed frank bilateral optic disc edema, confirmed by optical coherence tomography (OCT) (Fig. 1), with thickening of the peripapillary retinal nerve fiber layer (pRNFL) in all sectors of both optic discs.

Autofluorescence revealed a hyperautofluorescent image in both optic discs, corresponding to partially extruded drusen (Fig. 2). Cranial computed tomography (CT) performed in the emergency department and subsequent brain magnetic resonance imaging (MRI) excluded space-occupying lesions, and radiological signs of intracranial hypertension. Similarly, compressive or infiltrative etiology was ruled out, and no demyelinating lesions were observed. The only notable radiological finding was the presence of calcified bilateral optic disc drusen on the CT scan (Fig. 2).

During outpatient follow-up, laboratory tests including complete blood count, biochemistry, renal and hepatic profiles, erythrocyte sedimentation rate (ESR), CRP, proteinogram, immunoglobulins, vitamins, thyroid-stimulating hormone (TSH), angiotensin-converting enzyme (ACE), thrombophilia study, and autoimmunity markers (including anti-aquaporin-4 antibodies and anti-myelin-oligodendrocyte glycoprotein antibodies) all came out normal. Serology for syphilis, tuberculosis, toxoplasmosis, leptospirosis, *Bartonella*, *Mycoplasma*, *Borrelia burgdorferi*, hepatitis B and C viruses, HIV, and herpesvirus turned out negative. Visual evoked potentials showed a slight increase in bilateral conduction latency, more pronounced in OD.

Analysis of a semen sample by polymerase chain reaction detected *Chlamydia trachomatis* genetic material, leading to a therapeutic adjustment with the addition of a macrolide (azithromycin) to cover this agent. Since then, the patient has experienced a satisfactory progression: 1 month into targeted antibiotic therapy and resolution of prostatitis, he showed clear clinical improvement: VA was 1 in BE, fundus examination showed resolution of bilateral optic disc edema with persistence of the pseudopapilledema characteristic of optic disc drusen, computerized perimetry showed only peripheral defects, predominantly inferior (more marked in OD), and OCT showed thinning of the pRNFL, predominantly superior, again more marked in OD (Fig. 3). Two years after the resolution of the condition, the patient remains clinically, perimetrically, and tomographically stable.

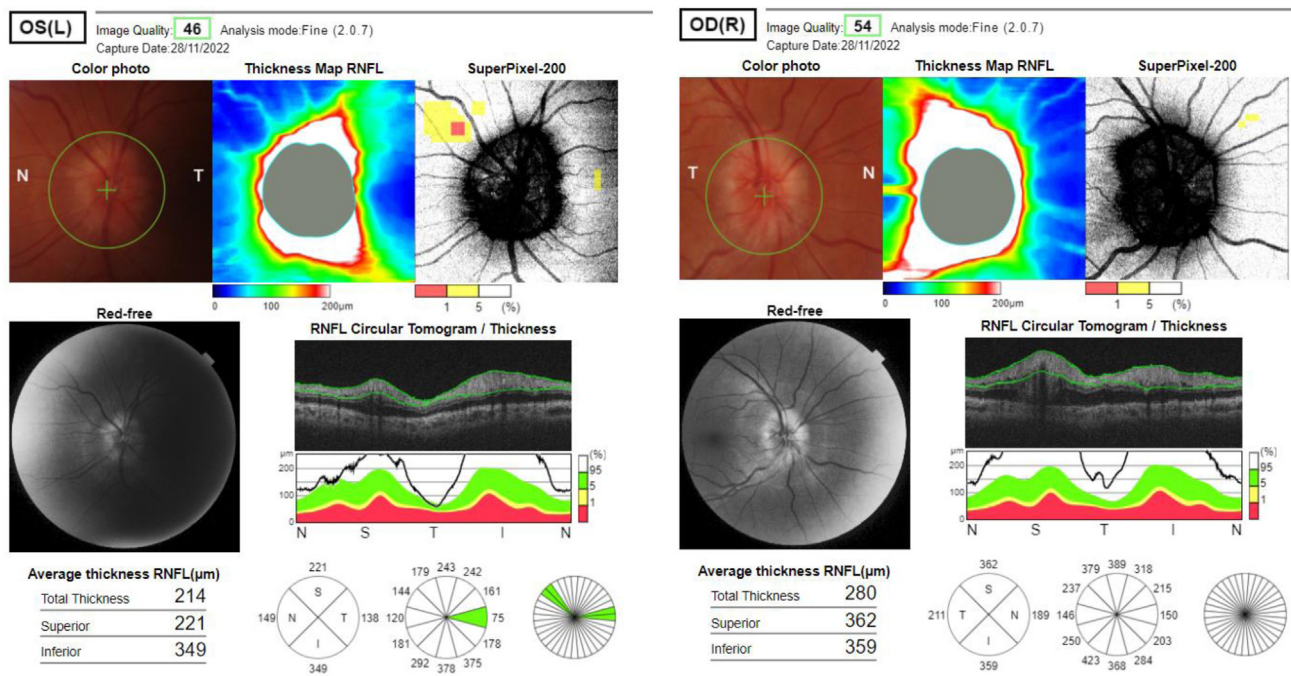


Fig. 1 – Bilateral optic disc edema (OCT and retinography).

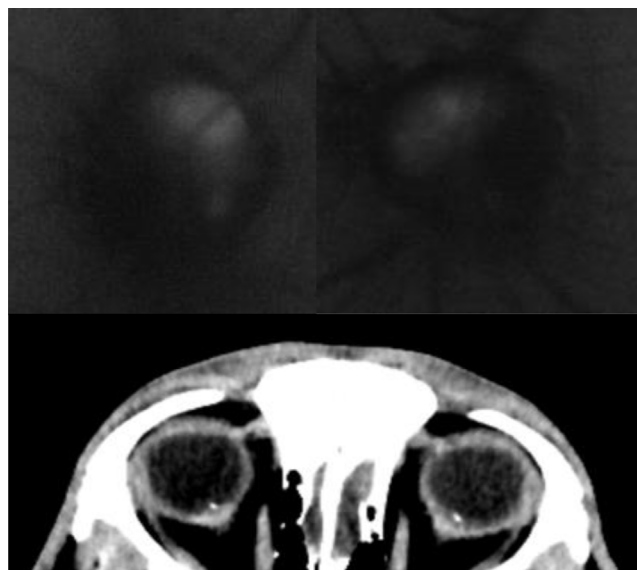


Fig. 2 – Optic disc drusen (autofluorescence, CT).

Discussion

Parainfectious optic neuritis is an optic neuropathy that occurs after a presumed or confirmed systemic infection.^{1,2} It is a rare entity generally associated with viral infections, with Epstein-Barr, varicella-zoster, rubella, and papovaviruses being the most common,^{1,2} although it has also been reported in bacterial infections such as those caused by *Mycoplasma pneumoniae*.^{1,2}

Since our patient presented with bilateral optic disc edema, we initially excluded both elevated blood pressure and a

space-occupying brain lesion. On the other hand, an unremarkable neurological examination, a brain MRI study without stigmas of ICH,⁶ and, especially, clear central visual involvement in OD, make papilledema unlikely.²

Since we are dealing with a young patient, without cardiovascular risk factors or recent surgical history and with negative autoimmunity markers, it is not considered, in principle, that the condition is due to ischemic optic neuropathy.²

Furthermore, the possibility of Leber's hereditary optic neuropathy (LHON) is not consistent with the over edema without telangiectatic microangiopathy of our patient's optic discs, nor to his visual involvement, as LHON presents with more frequently centrocecal and progressive defects.^{2,7}

In our case, the clinical presentation and laboratory, imaging, and neurophysiological studies suggest a bilateral inflammatory optic neuropathy.^{1,2} However, serology directed at infectious optic neuritis is negative, and similarly, both MRI findings and the autoimmunity study related to demyelinating optic neuritis are negative. Finally, the fact that the initial ophthalmological condition presented in the context of a systemic *C. trachomatis* infection and that its favorable progression with resolution of the infection was confirmed leads to the assumption of parainfectious optic neuritis as the most probable diagnosis.^{1,2}

Although the exact pathogenic mechanism of this subtype of neuritis is not completely understood, the fact that it appears days or weeks after the presumed systemic infection supports the theory that it is an autoimmune inflammatory reaction.⁸ Furthermore, *C. trachomatis* infection has been shown to induce a systemic immunomediated T-cell response,⁸ which could produce autoimmune damage; in addition, the release of pro-inflammatory mediators associated with the infection⁸ could indirectly affect the optic nerve.

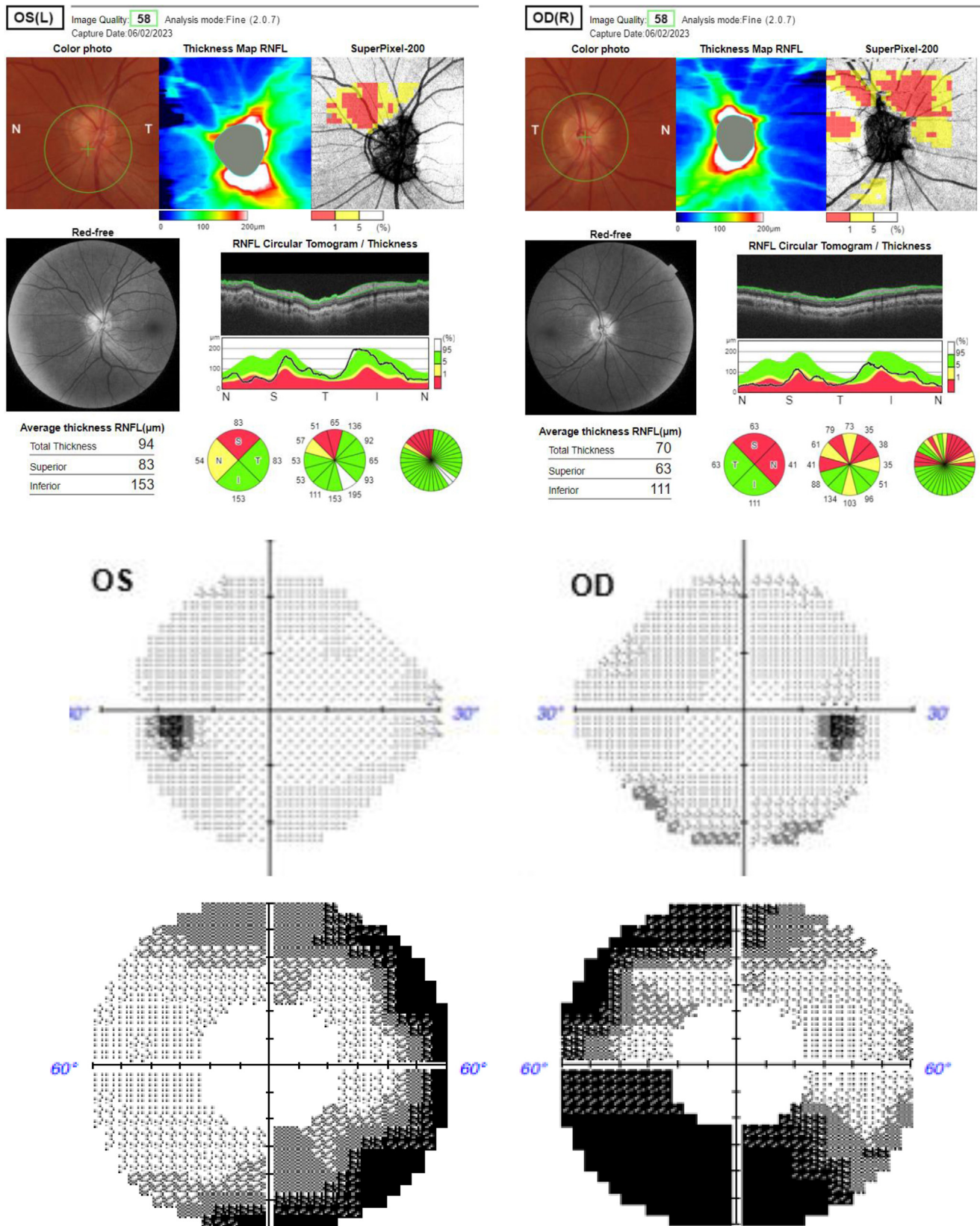


Fig. 3 – Resolution of edema and residual pRNFL atrophy (OCT and retinography), along with associated visual field defects (computerized perimetry).

Regarding treatment, although there is no clear consensus, systemic corticosteroids (IV pulsed methylprednisolone) are considered to accelerate recovery if there is severe bilateral vision involvement.^{1,2} In corticosteroid-resistant cases, plasmapheresis has been proposed as a therapeutic option.^{1,2} However, these interventions do not significantly alter the final visual prognosis, which is usually excellent once the initial infection has resolved.^{1,2} Thus, Vu et al.⁹ describe a case of parainfectious optic neuritis that showed complete visual recovery after antibiotic treatment only.

Our patient's visual impairment was not debilitating, so anti-inflammatory therapy was not considered, and he was treated with antibiotic therapy targeted at *C. trachomatis*.¹⁰ His final functional recovery was satisfactory, presenting only residual visual field defects that correlated with thinning of the pRNFL coinciding with the location of the extruded optic disc drusen (Figs. 2 and 3).

One of the main limitations in this case is the absence of clinical data prior to the papillitis episode, which makes it difficult to evaluate possible sequelae of the process and distinguish them from pre-existing damage caused by drusen extrusion. Furthermore, the infrequent nature of parainfectious optic neuritis^{1,2} and the scarcity of available studies prevent the establishment of more defined treatment protocols.

Finally, since we do not have a specific parameter that unequivocally establishes the causal relationship between a particular infection and the papillitis episode, it is possible that the responsible agent is an infectious agent other than *C. trachomatis* that went unnoticed.

In conclusion: parainfectious optic neuritis is a rare condition whose diagnosis requires a high index of suspicion and the exclusion of a wide variety of entities.^{1,2} In our case, it appeared associated with prostatitis caused by *C. trachomatis* and had the particularity of being superimposed on optic disc drusen.

Ethical considerations

The authors declare that they have adhered to evidence-based scientific clinical practice, and have obtained consent (verbal and written) from the patient to present the case while respecting their anonymity.

Funding

None declared.

Declaration of competing interest

None declared.

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